

TASK-EFFICIENT AGENT EVALUATION USING CACHED TRACES

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ABSTRACT

Efficient benchmarking estimates a system’s full-benchmark performance from a small number of task executions. Standard approaches rely solely on the binary correctness of executions from a collection of already-evaluated reference systems. An agentic system, however, produces a rich trace for each execution that can reveal substantially more about its capabilities. Exploiting this information requires useful trace representations. We propose representing a trace by extracting a structured, rubric-guided summary of it with an LLM and embedding the summary with a standard text encoder. These representations preserve correctness- and task-related signal while minimizing the presence of system identity-, model family-, and harness-related signals that dominate raw trace embeddings. Given this representation for each task execution from all reference systems, we construct low-dimensional representations of the systems, predict a new system’s score from its position among the reference systems, and combine the resulting geometric prediction with a two-parameter logistic item-response model. On SWE-bench Verified and Terminal-Bench 2.0, the proposed estimator reduces mean absolute error from 0.13 to 0.08 ($p < 10^{-4}$) and from 0.14 to 0.12 ($p < 10^{-3}$), respectively, when provided a single random executed task relative to an item-response model alone. These results indicate that trace geometry and correctness outcomes carry complementary predictive signal and enable more accurate benchmark score estimation from limited agent executions.

1 INTRODUCTION

The dominant paradigm of generative model use is increasingly agentic: systems plan, invoke tools, observe intermediate results, and iterate over long horizons to complete complex tasks (Yang et al., 2024; Wang et al., 2025). As agentic capabilities increase, so does the range of settings in which a given system might be deployed – and, with it, the number of benchmarks on which the system must be evaluated before it can be trusted with consequential work (Kapoor et al., 2026). Agentic evaluation, however, is expensive as each task execution requires a properly sandboxed environment and a single task execution routinely consumes hundreds of thousands of tokens (Jimenez et al., 2024; Terminal-Bench Team, 2026) and can regularly exceed \$2,000 (Kapoor et al., 2026). Given this cost, efficient evaluation is necessary to increase the evaluation coverage of an agentic system.

Task-efficient evaluation methods (e.g., benchmark compression and anchor-item selection (Vivek et al., 2024; Maia Polo et al., 2024), item response theory and adaptive testing (Kipnis et al., 2025; Truong et al., 2025), leveraging cached responses (Helm et al., 2026a), etc.) address this problem by finding structure in item-level information available from previously-evaluated reference systems to decrease the cost of evaluating a new system. Some task-efficient evaluation methods require only the item-level scores from reference systems (Vivek et al., 2024; Kipnis et al., 2025). Others leverage the content of cached artifacts, such as natural language responses or residual streams, to measure the similarity of systems on a set of tasks (Duderstadt et al., 2023; Helm et al., 2026b;a). Notably, agentic workflows often produce artifacts when executing a task – file diffs, tool calls, intermediate plans, and error messages – recorded together in an execution trace that can be used to this effect. Unfortunately, existing methods for using cached artifacts rely on off-the-shelf embedding functions that are not immediately well-suited for representing traces.

We make two primary contributions:

Table 1: Mean Absolute Error (MAE) of benchmark score estimators for SWE-bench Verified and Terminal-Bench 2.0 under leave-one-family-out evaluation and different task selection protocols (random & adaptive). **Bolded** values are the best method for a given benchmark, task selection, and number of tasks. Underlined values are statistically significant ($\alpha = 0.05$) comparisons to IRT per the two-sided Wilcoxon signed-rank test. The TRUBRIC and IRT blend – which takes advantage of both trace geometry and per-execution score – is the best or near-best in all settings.

Num. tasks m	SWE-bench Verified						Terminal-Bench 2.0					
	random			adaptive			random			adaptive		
	1	5	20	1	5	20	1	5	20	1	5	20
sample score	0.442	0.175	0.082	0.380	0.151	0.072	0.396	0.164	0.073	0.360	0.124	0.132
IRT	0.127	0.088	0.048	0.080	0.062	0.034	0.137	0.097	0.052	0.112	0.067	0.039
raw trace	<u>0.099</u>	0.093	0.089	0.096	0.083	0.081	0.135	0.126	0.120	0.128	0.125	0.113
generic	0.079	<u>0.071</u>	0.068	0.074	0.070	0.069	0.139	0.127	0.115	0.144	0.130	0.117
trubric	<u>0.081</u>	<u>0.069</u>	0.060	0.068	0.056	0.060	<u>0.127</u>	0.107	0.093	0.110	0.092	0.084
IRT + trubric	<u>0.081</u>	0.063	0.043	0.062	0.048	0.034	0.122	0.090	0.050	0.099	0.059	0.037

1. **TRUBRIC trace representations.** We introduce TRUBRIC (*trace rubric*): a trace representation constructed by extracting a task specific structured summary of each trace with an LLM and embedding the summary with a standard text encoder. We show that TRUBRIC embeddings preserve information relevant to the task and behavior of the system while minimizing system-, model family-, and harness- specific signal across seven modern text encoders.

2. **Task-efficient evaluation using cached traces.** Given the TRUBRIC embeddings from a set of tasks for a target system and a collection of reference systems, we evaluate their utility for predicting the target system’s full-benchmark score on SWE-bench Verified and Terminal-Bench 2.0 under three different agentic system evaluation protocols. We find that TRUBRIC-based estimators, combined with an item-response model, reduce single-task mean absolute error from 0.13 to 0.08 ($p < 10^{-4}$) and from 0.14 to 0.12 ($p < 10^{-3}$) on the two benchmarks, respectively, and are nearly always better than baselines at any task budget (Table 1).

1.1 PROBLEM STATEMENT

For our purposes, an agentic system $f \in \mathcal{F}$ is a random mapping from a task space $\mathcal{Q}$ to a trace space $\mathcal{X}$. A benchmark is a fixed collection of tasks $Q^* = \{q_1, \dots, q_M\} \subset \mathcal{Q}$. Each trace is evaluated by the benchmark’s trace-level scoring function $s : \mathcal{X} \rightarrow \{0, 1\}$ and the system is evaluated by its benchmark score $y(f) = \frac{1}{M} \sum_{j=1}^M \mathbb{E}[s(f(q_j))]$ where the expectation is with respect to the randomness of f . We write y for $y(f)$ when the context is clear.

When evaluating a new system, we typically have access to $\mathcal{D} = \{\{(x_{ij}, y_{ij})\}_{j=1}^M\}_{i=1}^n$, the traces and trace-level scores of n previously-evaluated reference systems $f_1, \dots, f_n$, where $x_{ij} \sim f_i(q_j)$, $y_{ij} = s(x_{ij})$ and $y_i = y(f_i)$. For the target system f_0 , we observe traces (and, optionally, trace-level scores) on only a subset $Q \subset Q^*$ of $|Q| = m \ll M$ tasks and our goal is to estimate the system’s full benchmark score y_0 using only the target’s m executions and $\mathcal{D}$.

2 METHOD

3 TRACE REPRESENTATIONS ENCODE CONFOUNDERS

4 EFFICIENT EVALUATION

5 RELATED WORK

Representing agentic traces. The standard tools for embedding model outputs are text encoders trained on sentence-level semantic similarity (Reimers & Gurevych, 2019; Su et al., 2023; Nussbaum & Duderstadt, 2025). Agentic traces, however, implicitly or explicitly encode system metadata that

may not be relevant for downstream tasks (Sun et al., 2025; Oderinwale, 2026) and are not necessarily encoded well by semantic embedding models. Instead, traces have primarily been studied through taxonomies of failure modes and actions (Cemri et al., 2025; Shu et al., 2026) and LLM judges of agent behavior (Zhuge et al., 2025; Mehtiyev & Assunção, 2026). Parallel work uses LLMs to adapt model outputs to be more naturally compared via their semantics (Ravfogel et al., 2024; Peng et al., 2024; Benara et al., 2024). TRUBRIC applies this to agentic traces by embedding a task-specific structured summary of each trace, preserving behavior-relevant information while minimizing the presence of system metadata.

Efficient benchmarking. Task-efficient benchmarking estimates a system’s full benchmark score from a small number of executions. The dominant paradigm for efficient benchmarking centers around item response theory (IRT) (Embretson & Reise, 2000), where a target system’s score is estimated with $m \leq M$ item-level scores by assigning a latent ability estimated from the relationship between a task’s difficulty and discriminability based on $\mathcal{D}$ (Maia Polo et al., 2024; Kipnis et al., 2025). Extensions of IRT select informative or adaptive task subsets (Vivek et al., 2024; Truong et al., 2025) or apply it to agentic benchmarks directly (Ge et al., 2026; Ndzomga, 2026). A complementary paradigm represents the systems themselves, either from their score profiles (Ruan et al., 2024; Zhuang et al., 2025) or from the artifacts they produce while executing a task (Duderstadt et al., 2023; Acharyya et al., 2024; Helm et al., 2025). For example, the Product Kernel Perspective Space (PKPS) (Anonymous, 2026) represents language models as Euclidean objects obtained via multidimensional scaling of the distances between their embedded responses. Helm et al. (2026a) regress benchmark scores on these representations and prove that the resulting predictions are task-efficient relative to the sample score. TRUBRIC enables extending geometry-based methods to agentic traces by providing an effective trace encoding. We show that these two approaches – score-based IRT and artifact-based PKPS – can be combined to outperform either for any number of given executed tasks.

6 DISCUSSION

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